



Current and Emerging Biomarkers for Sepsis: Diagnostic and Prognostic Tools

Kyeong Seob Shin^{†,*}

Department of Laboratory Medicine, Chungbuk National University College of Medicine, Cheongju 28644, Korea

ABSTRACT

Sepsis, a severe systemic inflammatory response to infection, poses a critical challenge in clinical practice due to its high morbidity and mortality rates. Despite advances in understanding sepsis pathophysiology, diagnosing the condition promptly remains challenging. This review explores current and emerging biomarkers used in sepsis diagnosis and management, highlighting their mechanisms, diagnostic performance, and clinical utility. Procalcitonin (PCT) and C-reactive protein (CRP) are among the most established biomarkers. PCT levels correlate with bacterial infection severity, aiding in early sepsis detection, guiding antibiotic therapy, and providing prognostic insights. CRP, though sensitive to inflammation, lacks specificity for bacterial infections, but it complements PCT in enhancing diagnostic accuracy. Additionally, cytokines such as interleukin-6 (IL-6) and tumor necrosis factor- α provide early indicators of sepsis, while IL-10 offers prognostic information on immune response regulation. Novel biomarkers like soluble triggering receptor expressed on myeloid cells-1 (sTREM-1), proadrenomedullin, heparin-binding protein, neutrophil gelatinase-associated lipocalin, and presepsin show potential in early sepsis diagnosis and risk stratification. Emerging biomarkers, including endotoxin activity assay, and TREM-2 promise to further refine sepsis diagnosis and prognosis. These markers, along with advanced technologies like genomics and proteomics, could improve diagnostic specificity and support personalized treatment strategies. By combining multi-biomarker panels with clinical and laboratory data, the review underscores the importance of integrated approaches in sepsis diagnosis and management, fostering better outcomes through precise, timely intervention.

Keywords: Sepsis, Biomarkers, Procalcitonin, C-reactive protein, Diagnosis

INTRODUCTION

Sepsis is a life-threatening condition characterized by

a widespread inflammatory response to infection, leading to tissue damage, organ failure, and potentially death, and despite advances in medical technology, it remains a global health concern with significant morbidity and mortality⁽¹⁾. Introduced in 1991, Sepsis-1 defined sepsis as a condition involving infection and systemic inflammatory response syndrome (SIRS) to identify early signs of sepsis⁽²⁾. However, because the SIRS criteria could be met in many non-sepsis conditions, there were concerns about overdiagnosis. To address this, Sepsis-2 included the category of severe sepsis with organ dysfunction, allowing

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[†]Corresponding author: Kyeong Seob Shin

Department of Laboratory Medicine, Chungbuk National University
College of Medicine, 1 Chungdae-ro, Seowon-gu, Cheongju 28644,
Korea

Tel: +82-43-269-6240, Fax: +82-43-271-5243

E-mail: ks shin@chungbuk.ac.kr

ORCID: <https://orcid.org/0000-0002-1680-1510>

*Professor.



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for a more accurate diagnosis and better identification of high-risk patients ⁽³⁾. In 2016, Sepsis-3 was introduced to provide a more precise and standardized approach to the diagnosis and management of sepsis, emphasizing the presence of life-threatening organ dysfunction caused by an unregulated host response to infection ⁽⁴⁾. Specifically, it uses the Sequential Organ Failure Assessment (SOFA) score to identify organ dysfunction, with an increase of 2 or more points in the qSOFA score indicating a potential sepsis diagnosis.

An ideal biomarker for sepsis should be highly sensitive and specific, easily measurable, and capable of providing real-time information about the patient's condition ^(5,6). Over recent years, multiple biomarkers have been proposed and investigated for their potential roles in sepsis management. These include traditional biomarkers such as C-reactive protein (CRP) and procalcitonin (PCT), along with newer candidates like presepsin, soluble triggering receptor expressed on myeloid cells-1 (sTREM-1), and various cytokines and chemokines. Given the complex interactions between the host immune system and invading pathogens in sepsis, it is unlikely that a single biomarker can capture the full spectrum of the disease. Consequently, recent studies have focused on integrated diagnostic approaches combining multi-biomarker panels with clinical evaluation and laboratory measurements ⁽⁶⁾.

This review provides a comprehensive examination of the current status of sepsis biomarkers, highlighting their biological significance, diagnostic performance, and clinical utility. It discusses the strengths and limitations of individual biomarkers and explores emerging trends in biomarker discovery and validation. Additionally, the potential of combining biomarkers with advanced technologies such as genomics, proteomics, and metabolomics to improve diagnostic and prognostic accuracy in sepsis will be explored.

RESULTS

1. Established biomarkers of sepsis

1) Procalcitonin

(1) Mechanisms of action

In sepsis, the rapid elevation of PCT levels is attributed to the host's systemic inflammatory response to bacterial infection. Under normal conditions, PCT is produced in the thyroid and converted to calcitonin, playing no significant role in inflammatory processes. However, during bacterial infections, inflammatory cytokines like tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6), along with bacterial endotoxins such as lipopolysaccharides, stimulate PCT production in multiple organs, bypassing thyroid regulation ⁽⁷⁾. This induction, primarily mediated by gene activation of the calcitonin gene-related peptide, results in a sharp increase in PCT that reflects the infection severity and overall inflammatory burden. Notably, PCT levels remain low during viral infections, highlighting its specificity for bacterial pathogens and distinguishing it from other markers of systemic inflammation, such as CRP ⁽⁸⁾.

(2) Clinical applications

The clinical relevance of PCT spans multiple aspects of sepsis management, including diagnosis, therapeutic monitoring, and prognostic evaluation. Elevated PCT levels serve as a specific indicator of bacterial infections, making it an effective marker for early sepsis detection, especially in distinguishing bacterial infections from viral or other non-infectious causes of inflammation. Clinically, PCT's half-life (about 24 hours) aligns well with infection dynamics, allowing for serial monitoring to assess therapeutic response and infection resolution. In the initial stages of sepsis, PCT levels rise rapidly, enabling early intervention. PCT's specificity for bacterial infections offers a diagnostic advantage over other inflammatory markers, reducing the likelihood of misdiagnosis and unnecessary treatment in cases where bacterial involvement is not present ⁽⁹⁾. Beyond its diagnostic utility, PCT play a significant role in antibiotic stewardship. By monitoring PCT levels, clini-

cians can make informed decisions about initiating or discontinuing antibiotics. For example, a declining PCT level indicates a positive response to treatment, suggesting the possibility of safely reducing or stopping antibiotic therapy. This practice helps minimize the duration of antibiotic exposure, reducing the risk of developing resistance and decreasing healthcare costs ⁽¹⁰⁾. Additionally, PCT provide valuable prognostic information, as its levels have been shown to correlate with disease severity and mortality risk in sepsis patients. Persistently high or increasing PCT levels despite treatment are associated with poorer outcomes and greater risk of complications. Thus, PCT can help stratify patients based on risk and inform decisions about escalation of care ⁽¹¹⁾.

(3) Diagnostic performance as a biomarker

Numerous studies affirm PCT's high diagnostic performance, with reported sensitivity ranging from 65%–75% and specificity from 70%–80% for bacterial sepsis ⁽⁹⁾. Although PCT alone provides strong diagnostic support, it is often used in conjunction with CRP and other markers to enhance diagnostic accuracy and improve patient outcomes. The combination of PCT with clinical assessments and other biomarkers yields a robust framework for early diagnosis, timely treatment, and dynamic monitoring of sepsis progression ⁽⁹⁾. Nevertheless, PCT may also rise in certain non-infectious conditions such as major trauma, surgery, or prolonged cardiogenic shock, which limits its absolute specificity for bacterial infection.

2) C-reactive protein

CRP is an acute-phase protein elevated in response to systemic inflammation, especially during sepsis, due to cytokine-driven hepatic synthesis. It aids in immune response by binding to pathogen surfaces, activating the complement system, and promoting opsonization. Clinically, CRP assists in diagnosing and monitoring sepsis progression, although it lacks specificity for bacterial infection. Its levels can guide antibiotic therapy and provide prognostic insights, but it is most effective when used

alongside other biomarkers, such as PCT, to enhance diagnostic accuracy in sepsis management.

(1) Mechanisms of action

CRP is an acute-phase protein predominantly synthesized by the liver in response to pro-inflammatory cytokines, particularly IL-6, TNF- α , and IL-1 β ⁽¹²⁾. During sepsis, the inflammatory response triggers hepatocytes to produce and release CRP into the bloodstream, leading to elevated CRP levels within hours of infection onset. This rapid elevation serves as a marker of systemic inflammation. CRP plays a role in the immune response by binding to phosphocholine on pathogen surfaces, promoting opsonization, and activating the complement system. Elevated CRP levels reflect systemic inflammation but are not specific to bacterial infection alone, as CRP can rise in other inflammatory conditions.

(2) Clinical applications

In sepsis management, CRP is commonly used for diagnosis, guiding antibiotic therapy, and assessing prognosis. As a diagnostic tool, CRP levels above 10 mg/L indicate a heightened inflammatory response, with levels often exceeding 100 mg/L in severe infections like sepsis ⁽¹³⁾. CRP can aid in early identification but lacks specificity for bacterial versus viral infections. In guiding antibiotic therapy, CRP can help clinicians monitor response to treatment; a reduction in CRP over 48–72 hours often suggests a favorable response, allowing for potential adjustments in therapy duration. Prognostically, persistently high or rising CRP levels are associated with poor outcomes and may indicate worsening sepsis or inadequate infection control ⁽¹⁴⁾.

(3) Diagnostic performance

While CRP is widely used due to its sensitivity to inflammation, its specificity in sepsis is limited. CRP's sensitivity generally exceeds 90% for detecting inflammatory conditions, including sepsis, but specificity is often lower (50%–70%) since it can elevate in non-bacterial infec-

tions and other non-infectious inflammatory states (e.g., trauma, surgery) ⁽¹³⁾. Compared to biomarkers like PCT, CRP lacks the ability to specifically distinguish bacterial infection in sepsis, though it provides valuable insight when used alongside clinical assessment and other biomarkers ⁽¹⁵⁾. Thus, CRP alone is insufficient for sepsis diagnosis, but it complements PCT and other biomarker to enhance diagnostic confidence and therapeutic monitoring.

3) Biomarker for sepsis: cytokines

Cytokines play a central role in the dysregulated host immune response during sepsis. Their rapid release and dynamic balance between pro- and anti-inflammatory mediators not only drive the pathophysiology of sepsis but also provide valuable insights for early diagnosis and prognosis. As such, several cytokines have been investigated as potential biomarkers in sepsis management ⁽⁵⁾.

(1) Interleukin-6

① Mechanism of action

IL-6 is a multifunctional cytokine that plays a pivotal role in the immune response during sepsis. It is produced by monocytes, macrophages, and endothelial cells in response to infection and tissue damage. IL-6 stimulates the liver to produce acute-phase proteins, including CRP, thus linking inflammation to systemic physiological responses. It also aids in lymphocyte differentiation and regulates the balance between pro-inflammatory and anti-inflammatory responses ⁽¹⁶⁾.

② Clinical application

IL-6 is widely used as a biomarker for early detection and monitoring of sepsis due to its rapid increase in response to infection. Elevated IL-6 levels are associated with increased severity of sepsis and poorer outcomes. Monitoring IL-6 levels over time can help clinicians assess the progression or resolution of sepsis, guiding treatment decisions and helping to identify high-risk patients ⁽¹⁷⁾.

③ Diagnostic performance

IL-6 shows high sensitivity in detecting sepsis, often

rising earlier than CRP or PCT, making it suitable for early diagnosis. However, because IL-6 can be elevated in a variety of inflammatory conditions, its specificity for sepsis is limited ⁽¹⁸⁾. To improve diagnostic accuracy, IL-6 is frequently used alongside other biomarkers, such as PCT or CRP, to enhance specificity in differentiating sepsis from other inflammatory conditions. However, since IL-6 can be elevated in autoimmune disease, trauma, or other systemic inflammatory states, its specificity is limited and it is best together with other markers such as PCT.

(2) Tumor necrosis factor- α

① Mechanism of action

TNF- α is a pro-inflammatory cytokine primarily produced by activated macrophages and monocytes. It initiates a cascade of inflammatory responses in sepsis, leading to increased vascular permeability, hypotension, and the recruitment of additional immune cells to the site of infection ⁽¹⁹⁾. TNF- α is one of the earliest cytokines released in response to infection, amplifying the inflammatory response and potentially leading to tissue injury and organ dysfunction in severe sepsis cases.

② Clinical application

TNF- α 's role in initiating the inflammatory cascade makes it a valuable marker for understanding the pathophysiology of sepsis. While TNF- α is not commonly used as a standalone diagnostic biomarker due to its short half-life and rapid clearance from circulation, it is crucial in research settings to elucidate early inflammatory changes. In clinical practice, TNF- α levels are often assessed alongside other cytokines to monitor the intensity of the inflammatory response in patients with severe sepsis or septic shock ⁽²⁰⁾.

③ Diagnostic performance

TNF- α levels rise very early in the course of sepsis, providing initial insight into the inflammatory status of the patient. However, its short half-life limits its routine use as a diagnostic marker, as levels can quickly return to baseline even in active sepsis. Additionally, TNF- α lacks specificity,

as it may be elevated in other inflammatory or infectious diseases. Thus, while valuable in research, TNF- α is rarely used as a primary diagnostic biomarker in clinical settings⁽²¹⁾. Due to its short half-life and lack of specificity, TNF- α is rarely adopted in routine clinical practice and remains primarily of research interest.

(3) Interleukin-8

① Mechanism of action

IL-8 is a chemokine that plays a crucial role in neutrophil activation and recruitment to the site of infection. During sepsis, IL-8 levels surge, guiding neutrophils to the infection site, where they contribute to pathogen clearance⁽²²⁾. This recruitment of immune cells is essential for containing and eliminating infection, but an excessive IL-8 response can lead to an overwhelming inflammatory response, contributing to tissue damage.

② Clinical application

IL-8 is valuable in assessing the immune response's severity and helps predict outcomes in sepsis patients. Elevated IL-8 levels are often correlated with the severity of sepsis, and high levels are linked with poorer outcomes, such as organ dysfunction or mortality. Measuring IL-8 can be helpful in triaging sepsis patients and monitoring response to treatment, particularly in critically ill patients⁽²³⁾.

③ Diagnostic performance

IL-8 shows promise as a biomarker due to its sensitivity to bacterial infections and its rapid response in early sepsis stages. However, like TNF- α , IL-8 can be elevated in other inflammatory conditions, which limits its specificity⁽²⁴⁾. Therefore, IL-8 is often measured alongside other markers, such as PCT or CRP, to improve diagnostic accuracy and to differentiate between sepsis and other inflammatory states.

(4) Interleukin-10

① Mechanism of action

IL-10 is an anti-inflammatory cytokine produced by regulatory T-cells, macrophages, and monocytes in response

to inflammation. In sepsis, IL-10 acts as a counter-regulatory mechanism to mitigate the pro-inflammatory response, helping to prevent excessive tissue damage⁽²⁵⁾. It downregulates the production of pro-inflammatory cytokines, such as TNF- α and IL-1, contributing to immune homeostasis. However, persistently elevated IL-10 levels can indicate immune suppression, which may increase susceptibility to secondary infections and worsen sepsis outcomes.

② Clinical application

IL-10 is often used as a prognostic marker rather than a diagnostic one, as high levels are associated with an increased risk of mortality and immune dysregulation in sepsis patients⁽²⁶⁾. Monitoring IL-10 can provide insights into the balance between pro- and anti-inflammatory responses, helping clinicians to assess the immune status of patients. Elevated IL-10 may also signal the need for closer monitoring and more aggressive therapeutic interventions.

③ Diagnostic performance

IL-10's role as an anti-inflammatory cytokine makes it less useful for diagnosing sepsis compared to pro-inflammatory markers. However, IL-10 is a valuable prognostic marker; elevated levels often correlate with poor prognosis and can indicate an overcompensated immune response, particularly in severe cases of sepsis⁽²⁷⁾. As a biomarker, IL-10 is most informative when used in conjunction with pro-inflammatory markers like IL-6, providing a more comprehensive picture of the immune response in sepsis. IL-10 is there better regarded as a prognostic marker of immune dysregulation rather than a diagnostic tool for sepsis.

The representative established biomarkers and their characteristics are summarized in Table 1^(7,9,12,13,16,17,19,22–26).

2. Emerging and clinically promising biomarkers for sepsis

Despite the utility of established biomarkers such as CRP and PCT, their limited specificity and sensitivity high-

Table 1. Established sepsis biomarkers

Biomarker	Mechanism	Clinical utility	Diagnostic performance	References
PCT	Induced by bacterial endotoxins and cytokines (TNF- α , IL-6)	Early diagnosis, antibiotic guidance, prognosis	Sensitivity 65%–75%, specificity 70%–80%	(7,9)
CRP	Hepatic synthesis stimulated by IL-6, TNF- α	Detects inflammation, monitoring; limited specificity	Sensitivity >90%, specificity 50%–70%	(12,13)
IL-6	Produced by immune cells, induces acute-phase proteins	Early indicator, severity marker, prognosis	High sensitivity; limited specificity	(16,17)
TNF- α	Early pro-inflammatory cytokine	Research use; reflects inflammation intensity	Short half-life; not routine	(19)
IL-8	Chemokine recruiting and activating neutrophils at infection site	Reflects severity; elevated levels linked to poor outcomes; prognosis and patient stratification	Early rise in bacterial infection; correlates with severity; limited specificity	(22-24)
IL-10	Anti-inflammatory cytokine	Prognostic indicator for immune suppression	Not diagnostic	(25,26)

PCT, procalcitonin; CRP, C-reactive protein; IL, interleukin; TNF- α , tumor necrosis factor- α .

light the need for novel markers. Recent studies have therefore focused on emerging biomarkers that can provide earlier detection, better prognostic information, and more precise risk stratification in sepsis. These promising candidates include sTREM-1, proadrenomedullin (ProADM), and heparin-binding protein (HBP), among others ^(5,6).

1) Soluble triggering receptor expressed on myeloid cells-1

(1) Mechanism of action

sTREM-1 is expressed on the surface of neutrophils and monocytes, becoming activated in response to infection. When an infection is present, sTREM-1 is released into the bloodstream, amplifying the inflammatory response. It plays a key role in modulating immune responses in sepsis by promoting cell-mediated immune reactions and inducing cytokine release ⁽²⁸⁾.

(2) Clinical utility

sTREM-1 is valuable for differentiating the presence of infection and assessing the severity of the inflammatory response. It aids in early diagnosis of sepsis by helping to distinguish infectious sepsis from non-infectious inflammatory conditions.

(3) Diagnostic performance

sTREM-1 demonstrates high sensitivity and specific-

ity, making it a promising biomarker for sepsis diagnosis. Supporting evidence indicates that sTREM-1 is specifically upregulated in bacterial infection, while sterile inflammation such as trauma or postoperative stress produces minimal activation, which underlies its ability to distinguish infectious from non-infectious conditions. However, it is recommended to use it alongside other biomarkers (such as PCT and CRP) to improve diagnostic accuracy.

2) Proadrenomedullin

(1) Mechanism of action

ProADM is the precursor of adrenomedullin, a hormone essential for vascular regulation and inflammatory response. During sepsis, ProADM levels rise due to increased vascular permeability and excessive inflammatory activation ⁽²⁹⁾.

(2) Clinical utility

ProADM reflects endothelial function and overall hemodynamic status, making it useful for prognosis and severity assessment in sepsis patients. Elevated ProADM levels in acute infections are associated with worse clinical outcomes.

(3) Diagnostic performance

ProADM is better suited for prognosis rather than diagnosis in sepsis. However, when used alongside other bio-

markers, it can play a supportive role in early diagnosis and severity assessment. Although not specific for infection, ProADM provides incremental value by reflecting endothelial dysfunction and circulatory stress, which are central features of severe sepsis and septic shock.

3) Heparin-binding protein

(1) Mechanism of action

HBP is a protein released by activated neutrophils, significantly increasing vascular permeability. During inflammation, it facilitates protein translocation across vascular walls, contributing to endothelial damage and the development of shock ⁽³⁰⁾.

(2) Clinical utility

HBP is valuable for early prediction of septic shock, particularly for detecting severe circulatory disturbances related to sepsis. HBP levels increase early in infection, providing crucial information for rapid sepsis diagnosis and treatment.

(3) Diagnostic performance

HBP is highly predictive of septic shock, showing high diagnostic performance in this context. Elevated HBP levels in patients with suspected infection signal a higher risk for septic shock, proving its usefulness in early risk assessment. Its ability to predict progression to septic shock makes HBP particularly useful in the emergency setting for early triage of high-risk patients.

4) Neutrophil gelatinase-associated lipocalin

(1) Mechanism of action

Neutrophil gelatinase-associated lipocalin (NGAL) is a protein secreted by neutrophils and renal tubular epithelial cells. It rapidly increases during acute kidney injury (AKI) and reflects neutrophil activation in infection and inflammation ⁽³¹⁾.

(2) Clinical utility

NGAL is most valuable for early detection of AKI in sepsis, often rising 24–48 hours before creatinine. It aids in

assessing sepsis severity and complications, particularly renal dysfunction, and supports risk stratification in patients admitted to the intensive care unit ⁽³²⁾.

(3) Diagnostic performance

For AKI detection, NGAL demonstrates an area under the curve (AUC) of approximately 0.75–0.85. While less specific for diagnosing sepsis itself, NGAL provides high clinical value for predicting and monitoring sepsis-associated complications ⁽³³⁾.

5) Presepsin (sCD14-ST)

(1) Mechanism of action

Presepsin is a soluble fragment of CD14, released during activation of monocytes and macrophages in response to bacterial infection. It reflects early immune activation, particularly through the LPS-CD14-TLR4 signaling pathway ⁽³⁴⁾.

(2) Clinical utility

Presepsin is valuable for early diagnosis of bacterial sepsis, often rising more rapidly than PCT. It also supports severity classification (distinguishing sepsis from septic shock) and has shown usefulness in neonatal and pediatric sepsis ⁽³⁵⁾.

(3) Diagnostic performance

Meta-analyses have reported an AUC of approximately 0.80–0.85, with sensitivity and specificity comparable to or exceeding those of PCT in some studies. Presepsin is particularly strong in early detection and prognostic assessment ⁽³⁶⁾. Nevertheless, larger multicenter studies are still required to validate its cutoff values and standardize its clinical use.

A summary of these emerging and clinically promising biomarkers is provided in Table 2 ^(28–30,32,33,35,36).

3. Emerging and additional sepsis biomarkers

While biomarkers such as CRP, PCT, sTREM-1, ProADM, and HBP are widely studied and have shown clinical relevance, their limitations in specificity, sensitivity, and

Table 2. Emerging and clinically promising biomarkers

Biomarker	Mechanism	Clinical utility	Diagnostic performance	References
sTREM-1	Released by neutrophils/monocytes during infection	Early infection detection, risk stratification	High sensitivity and specificity (with others)	(28)
ProADM	Precursor of adrenomedullin, indicates vascular stress	Prognosis, hemodynamic status	High prognostic value	(29)
HBP	From neutrophils; increases vascular permeability	Early predictor of septic shock	Highly predictive for shock	(30)
Presepsin	Fragment of CD14; monocyte activation marker	Early detection, severity grading	Promising; under validation	(35,36)
NGAL	From kidney; marker of injury	Early AKI detection in sepsis	Good for AKI presence	(32,33)

sTREM-1, soluble triggering receptor expressed on myeloid cells-1; ProADM, proadrenomedullin; HBP, heparin-binding protein; NGAL, neutrophil gelatinase-associated lipocalin; AKI, acute kidney injury.

applicability across diverse patient populations necessitate the search for additional markers. Recent advances in immunology, genomics, and proteomics have enabled the discovery of novel candidates that may capture different aspects of the host–pathogen interaction, including immune dysregulation, endothelial dysfunction, and organ injury. These emerging and additional biomarkers hold promise not only for improving diagnostic accuracy but also for providing deeper insights into the pathophysiology of sepsis and guiding personalized therapeutic strategies. Below are a few noteworthy examples that are currently under investigation^(5,6,23).

1) Endotoxin activity assay

Endotoxin activity assay (EAA) is designed to measure the presence and activity of endotoxins, which are bacterial components that trigger intense immune responses in sepsis. By quantifying endotoxin activity, EAA provides a real-time indication of bacterial load and endotoxemia, especially useful in Gram-negative sepsis cases. Its specificity for endotoxemia makes it valuable for tailored therapeutic approaches, though it is not universally applicable across all infection types⁽³⁷⁾. EAA is therefore particularly useful in Gram-negative sepsis, where endotoxemia is a defining feature, and may help guide targeted therapeutic interventions.

2) Triggering receptor expressed on myeloid cells-2

Like sTREM-1, TREM-2 is another receptor involved

in immune responses. TREM-2 expression is associated with modulating anti-inflammatory processes and tissue repair, representing a novel approach to assess the inflammatory balance in sepsis. TREM-2 may be valuable for determining sepsis severity, as low levels are associated with poor outcomes⁽³⁸⁾. Its downregulation in severe sepsis has been associated with poor outcomes, suggesting potential as a prognostic biomarker especially in immunosuppressed populations.

3) C-X-C motif chemokine ligand 10

C-X-C motif chemokine ligand 10 (CXCL10) is a chemokine induced by interferon- γ during immune responses to infection. Elevated CXCL10 levels are linked with heightened immune activation, and it is currently investigated as a potential biomarker for predicting the progression to severe sepsis and organ dysfunction⁽³⁹⁾. Because CXCL10 is also elevated in viral infections, its specificity for bacterial sepsis is limited; however, it retains value as a prognostic marker for progression to organ dysfunction.

4) Pentraxin 3

Pentraxin 3 (PTX3), a member of the pentraxin family alongside CRP, is rapidly released during infection and inflammation. It offers greater specificity for infections than CRP and is being explored for its potential in predicting sepsis outcomes and organ dysfunction more accurately than traditional markers⁽⁴⁰⁾. Compared with CRP, PTX3

Table 3. Experimental or investigational biomarkers

Biomarker	Mechanism	Clinical utility	Diagnostic performance	References
CXCL10	Induced by IFN- γ ; immune activation marker	Potential for severity prediction	Under evaluation	(39)
TREM-2	Modulates anti-inflammatory/tissue repair	Immune balance; low levels associated with poor outcomes	Investigational	(38)
PTX3	Acute-phase protein like CRP, more specific	Infection/organ dysfunction marker	Possibly better specificity than CRP	(40)
EAA	Measures endotoxin activity	Gram-negative sepsis, targeted therapy	Specific for endotoxemia	(37)

CXCL10, C-X-C motif chemokine ligand 10; TREM-2, triggering receptor expressed on myeloid cells-2; PTX3, pentraxin 3; EAA, endotoxin activity assay; CRP, C-reactive protein.

shows greater infection specificity and is under investigation as a candidate to improve prognostic accuracy in sepsis.

Additional investigational biomarkers currently under evaluation are listed in Table 3^(37–40).

CONCLUSION

Sepsis remains a major global health burden, and despite decades of research, its timely diagnosis and precise management remain challenging. Established biomarkers such as PCT and CRP are widely used in clinical practice, but their limited specificity and sensitivity underscore the need for additional tools. Cytokines including IL-6, TNF- α , and IL-10 provide valuable insights into the host immune response and disease severity, while emerging candidates such as sTREM-1, ProADM, and HBP show promise for improving early recognition and risk stratification. Nevertheless, no single biomarker is sufficient to capture the complex pathophysiology of sepsis, which involves dysregulated inflammation, immune suppression, endothelial dysfunction, and multi-organ injury. Future research should therefore prioritize integrated approaches that combine multiple biomarkers with clinical scoring system, imaging modalities, and advanced molecular technologies. Multi-marker panels, when interpreted in the context of patient characteristics and comorbidities, may enhance diagnostic accuracy, optimize therapeutic decisions such as antibiotic stewardship, and facilitate individualized prognostic assessment. Moreover, advances in high-

throughput genomics, proteomics, and metabolomics are expected to identify novel biomarkers and generate personalized sepsis signatures, enabling earlier intervention and reducing mortality.

In conclusion, while significant progress has been made in the identification and validation of sepsis biomarkers, successful translation into routine practice will require rigorous validation in diverse populations, standardization of cutoff values, and incorporation into clinical algorithms. A concerted effort among researchers, clinicians, and policymakers will be essential to move from discovery to bedside application, ensuring that biomarker-driven approaches meaningfully improve sepsis diagnosis, management, and prognosis.

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ORCID

Kyeong Seob Shin, <https://orcid.org/0000-0002-1680-1510>

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